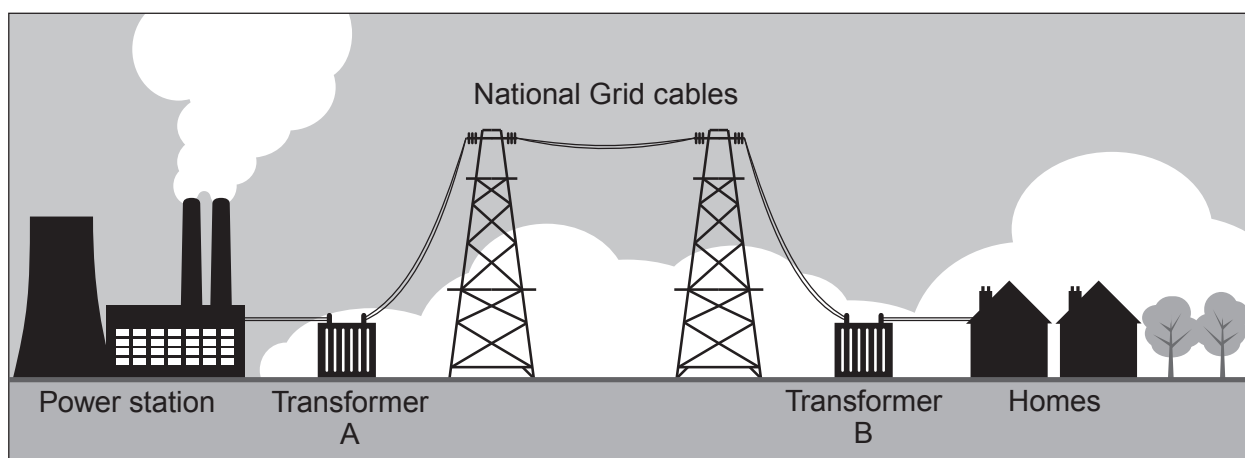


5. The National Grid includes a large network of cables that connects power stations to homes.



(a) Describe the purpose of transformers A and B in the National Grid.

Include in your answer:

- how electrical energy is transmitted **efficiently**
- how electrical energy is transferred **safely** to homes.

[6 QER]



- (b) Some houses have solar panels on their roof.
They produce electricity.
Explain why these houses are still connected to the National Grid. [2]

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.....

.....

- (c) A housing development of 20 new houses needs to be connected to the National Grid.
Each house requires a **maximum current of 60 A**.

- (i) Calculate the total current needed if all 20 houses are using the maximum current. [1]

Total current = A

- (ii) The mains voltage in the houses is 230 V a.c.

- I. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the maximum total power supplied to the 20 houses. [2]

Maximum power = W

- II. Convert your answer for the maximum power into kW. [1]

Maximum power = kW

